CausalLayer Engine v1.3.0 — Release Report

Author: Manus AI **Date:** May 15, 2026

Executive Summary

The v1.3.0 engine cycle successfully closed the generalisation gap exposed by the v0.8 corpus expansion. By hardening the engine against four distinct attack surfaces (attorney sanctions, defamation-by-AI, aviation sensor handovers, and algorithmic government action), we recovered the vast majority of the fresh-case misses while preserving baseline stability.

Headline numbers (CALB-2 v0.8.0, 250 cases):

- **Legal Top-1, all cases:** 225 / 250 = **90.0 %** (up from 86.0 % in v1.2)
- **Legal Top-1, high-confidence:** 163 / 175 = **93.1 %** (up from 88.6 % in v1.2)
- **Fresh-case accuracy:** 41 / 48 = **85.4 %** (up from 62.5 % in v1.2)
- **Baseline stability:** 184 / 202 = **91.1 %** (negligible −0.5 pp drift from v1.2)

The Generalisation Gap Closure

The v0.8 expansion introduced 48 fresh cases that severely tested the engine's ability to generalise to new doctrine surfaces. v1.2.0 struggled, dropping to 62.5% on the fresh cohort. v1.3.0 closes this gap almost entirely.

Generalisation Gap Closure

What Landed in v1.3.0

Attack Surface	Doctrine Refinement	Cases Unblocked
FRCP-11 Attorney Sanctions	Removed the strict `humanInTheLoop` telemetry gate. The rule now fires reliably on `legal_services` + `hallucination_fabrication` + court/disciplinary sources, reflecting the non-delegable duty of candour.	L7-201 (Mata), L7-202 (Park), L7-203 (Wadsworth), L7-204 (Crabill), L7-205 (DPP v AB)
Defamation by AI / § 230	Added a dedicated Publisher Liability rule that intercepts `defamation_by_ai` cases before they hit Strict Product Liability. Routes to the human prompter (if named) or the affected party when courts dismiss under Section 230.	L7-251 (Walters v OpenAI)
Aviation Sensor Handovers	Broadened the Level-2 detector to recognize global aviation safety boards (EAIB, KNKT, TSB, etc.) and `perception_sensor_failure`. Added specific apportionment branches for manufacturer design defects (Boeing MCAS) and multi-causal airline operations (SJ-182).	L9-203 (ET302), L9-205 (SJ-182), L4-205 (Waymo Recall)
Algorithmic Government Action	Fixed a routing bug where in-house government algorithms were being preempted by the Regulatory Action rule. The Government Administrative Liability doctrine now correctly fires first for these cases.	L8-201 (Robodebt), L8-204 (Allegheny)

Cryptographic Anchor

The v1.3.0 run over the v0.8.0 corpus has been anchored to the Bitcoin blockchain via OpenTimestamps.

- **Merkle Root:** `51576e5a4b30c7e407ff099d5dd07d7e1960ef0a24fa30e9eb6fb9d487c3cc1b`
- **Leaf Count:** 250

- **File:** `causallayer-anchor-log/anchors/2026-05-15-v1.3-calb2-v0.8.0.json(.ots)`

Known Limitations & v2.0 Architectural Debt

The engine’s `PartyAttribution` type is strictly limited to four slots (`ai_provider`, `deployer`, `data_provider`, `human_operator`). It cannot express `external_actor` or `affected_party`.

This structural limitation makes cases like L1-256 (*Battle v. Microsoft*, GT=`external_actor=100`) and L4-204 (*NTSB Williston*, GT=`external_actor=100`) permanently unscorable. Expanding the attribution vocabulary to all six slots is the primary architectural requirement for v2.0.